Food and Power: Agricultural Policy under Democracy and Dictatorship

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Government distortions to agricultural markets have long been recognized as a key determinant of economic growth and development, and a major factor affecting global trade.[1] However, political scientists do not have a complete understanding of how these agricultural policies are made. Specifically, although we know that government market interventions tend to lower food prices and decrease the incomes of farmers in poorer and less democratic countries, research has not explored how agricultural policy varies among authoritarian regimes.[2]

In this article, I argue that institutions and socio-economic structures interact to generate agricultural policy outcomes. Because authoritarian governments are less responsive to electoral incentives and thus to the interests of the rural population, they implement urban-biased policies that decrease returns to farmers compared to democracies, on average. However, higher levels of agricultural support occur under autocracy when landholding inequality creates a small, powerful landed elite. Income inequality generates redistributive pressure under democracy and hampers food consumers' ability to mobilize against authoritarian regimes, also leading to relatively high levels of support for agriculture among autocracies. When urban interests are powerful, autocracies provide lower levels of agricultural support than do democracies, which also implies lower food prices. I estimate cross-national panel regressions for fifty-six countries between 1960 and 2010, finding significant variation in policy outcomes across democracies and dictatorships which confirms these theoretical predictions.

This is a significant step towards explaining the sizeable variation in policy both across regime types and among authoritarian regimes. In 1995, for example, the average authoritarian regime decreased agricultural producer prices by around 3 percent vis-à-vis world market prices, while the average democracy increased them by 29 percent. However, the greatest level of support for agriculture under autocracy was seen in Russia, which increased producer prices by 24 percent, while the lowest was observed in Sudan, which decreased prices by 32 percent.[3] Explaining this variation looms as an important task when we consider that the average authoritarian regime has a significantly greater share of employment and output in the agricultural sector than the average democracy while, at the same time, the average citizen under an authoritarian regime spends a significantly larger proportion of their income on food.[4] Agricultural market distortions, with their direct impacts on farmers' incomes and consumers' living standards, are among the most economic and politically salient policy areas in developing autocracies.

This article also contributes to an emerging field of scholarship that investigates interest aggregation and economic policy under authoritarianism. In contrast to previous work, I construct a theory of authoritarian agricultural policy as a response not only to the size of the selectorate, but to the threats of powerful economic actors.[5] Neither agricultural producers nor food consumers are small special interest groups in autocracies. Existing theories struggle to predict how a regime with a selectorate of a given size—that is, with a certain level of democratic accountability—would manage their directly opposed, yet diffuse, interests. In my account, explaining whether policy follows the preferences of one group or the other requires assessing the political threats of each and how they are addressed by the regime.

Existing Literature: Agricultural and Economic Policy-Making under Democracy and Dictatorship

This article addresses an important gap in existing research on the political economy of agricultural policy and the political economy of authoritarian regimes. Although previous work has shown that governments tend to support farmers more as their economies develop and as they become more democratic, very little research has examined variation in the agricultural policy environment among authoritarian regimes. Studies have measured the effects of the various policies put in place by governments to distort domestic agricultural markets, which range from trade measures such as export tariffs through subsidies for inputs to food price ceilings and transfer payments. These data have consistently revealed the trend for government support for farmers to increase as countries develop and the importance of agriculture in their economies declines.[6]

Non-democratic regimes have been found to be less responsive to the interests of rural agricultural producers and have lower levels of producer support and lower consumer food prices than democracies, on average. Authoritarian governments follow urban-biased agricultural policies, as they discriminate against the rural population in order to cheapen food and non-farm inputs, promote industrialization, and enrich urban and commercial interests.[7] Wallace argues that urban-biased policies are "overdetermined and endemic in poor nondemocracies" as authoritarian governments implement policies which decrease food prices to prevent urban unrest in the short term.[8] Democratic governments, holding all else equal, are significantly less likely to discriminate against the agricultural sector because they need the support of rural voters in order to win elections. Olper and Raimondi, for example, show that democratic transitions are associated with increases in support for farmers.[9] Similarly, Bates and Block find that improvements in executive accountability in Africa since the 1960s have been associated with increased price supports in the region.[10]

This relatively large body of literature on policy differences between democracies and dictatorships does not account for significant differences in agricultural policy-making and policy outcomes among dictatorships. Although Bates identified several groups able to extract relatively favorable policies from authoritarian governments—such as the rice industry in Ghana and the large-scale farmers of Kenya—these are rare exceptions to the general rule of urban-biased policies under authoritarianism outlined above.[11] No work has attempted a systematic theory which seeks to explain agricultural policy variation under authoritarian regimes and lay out the conditions under which policy outcomes do and do not differ from democracy to dictatorship.

This gap in the literature is surprising when one considers the importance of the agricultural sector in the average economy ruled by authoritarian regimes and the significant role played by these countries in global agricultural trade. It is, however, not surprising when one considers the relative paucity of scholarship examining variation in economic policy among authoritarian governments. Research has found that nondemocracies spend less on primary education and health than democracies and are less effective in translating economic growth into positive social outcomes such as lower child mortality rates.[12] However, we know far less about differences in economic policy outcomes among nondemocracies, and, according to one prominent account, dictatorships are "too much of a mixed bag" to attempt any general theory of their policies.[13]

Theories of authoritarian rule and democratization generate only general predictions on the balance between repressive and redistributive policies and are not applicable to conflicts within a single sector like agriculture.[14] Previous research on economic policy under authoritarianism has focused on the size of the "selectorate" rather than its composition as an explanation of policy variation. Along these lines, Steinberg and Malhotra find that regimes with smaller selectorates are more responsive to selective interests and more likely to follow a fixed exchange rate.[15] Relatedly, Hankla and Kuthy find that more institutionalized regimes follow open trade policies because they are less likely to favor selective interests, while Wright argues that binding legislatures constrain confiscatory behavior and foster economic growth.[16]

Here, I push this research agenda forward in two ways. First, I extend it to explain agricultural market distortions, a policy area of great importance for development and international trade. Second, I do not base my analysis on selectorate theory and the institutional characteristics of authoritarian regimes. Rather than examining the size of the selectorate, my approach looks to socio-economic factors to derive testable hypotheses on the relative political and economic power of competing groups—rural and urban interests—within the agricultural sector. I predict that authoritarian governments seeking survival must respond to the interests of groups who can plausibly threaten their position of power, even if they are relatively diffuse.[17] My power-based approach to economic policy-making under authoritarianism is not without precedent: recent papers by Pierskalla and Wu also highlight the role of more diffuse threat structures in economic policy-making under dictatorship.[18] However, my focus on mass, rather than elite, interests is a relatively novel and important theoretical addition to existing institutional approaches.

Explaining Agricultural Policy Outcomes under Democracy and Dictatorship

My argument is based on the premise that authoritarian governments formulate agricultural policy to address threats from powerful interests. They do so in a fundamentally different way than democratic governments, responding to the power of competing groups to organize in collective action against a regime rather than electoral incentives. The power of competing interests to threaten a regime is based, in turn, on structural features of the economy. In my empirical analysis, I therefore use socio-economic variables as measures of consumer and producer power and examine their effects on policy outcomes under democracy and dictatorship.

Democratic governments respond to electoral incentives: they weigh up the interests and political support of producers and consumers in terms of campaign contributions and votes, respectively, when making economic policy decisions.[19] On the other hand, authoritarian leaders respond to a different type of incentive: the threat of regime instability and being removed from office by powerful economic interests. They are concerned with promoting the interests of economic actors who are key supporters of, or threats to, their regime. Their overriding goal in following the interests of powerful actors is securing their position in power and preventing regime instability.[20] As Steinberg and Shih show in their study of Chinese trade policy, for example, even in states with a high degree of autonomy, leaders make economic policies that are supported by powerful domestic interest groups in order to ensure their political survival.[21] Wallace applies a similar logic to agricultural policies, which he argues are a function of political expediency on the part of authoritarian governments seeking to avoid unrest among urban food consumers.[22] I argue that this is only one side of the story: autocratic governments use agricultural policy not only to address threats from urban food consumers but also those from rural food producers.

In a political system where power is not allocated through elections, the relative threat posed to a regime by competing groups depends on the economic resources at each group's disposal and their ability to organize in collective opposition to the government—what I call their political power. In the case of food consumers, I draw on insights from existing work on the determinants of regime collapse and in development economics to show that consumers' political power is greater when the population is concentrated in cities and diminished when income inequality is high. In the case of agricultural producers, I build upon existing work in comparative politics to show that farmers' political power under authoritarian governments is greater when landholding inequality is high.

Urban interests form a powerful constituency when urbanization is high under autocracy, while the geographic distribution of the non-agricultural population does not decisively affect electoral politics under democracy, leading to lower levels of agricultural support under more urbanized autocracies compared to democracies.[23] The power of concentrated urban interests derives from both the mass and elite levels under autocracy. The mass of urban residents are politically active, influential, and very sensitive to changes in their real incomes, especially those due to shifts in food prices as they cannot produce any food themselves. Food makes up a large proportion of many urban dwellers' budgets in low-income dictatorships, and when this segment of the population airs its grievances over standards of living, they often center on food affordability. Because they are concentrated in a relatively small space, urban consumers find it easier to solve collective action problems and quickly mobilize in defense of their economic interests, for example to protest against government policies which result in high food prices.[24] Under authoritarian governments, where electoral accountability is lower than in democracies and unrest is more common, consumers' ability to organize in collective opposition to policies that increase food prices gives them a relatively high degree of political power. This is especially true in comparison to rural populations, which are typically poorer than city dwellers and find it more difficult to organize in direct opposition to the government.[25] However, aside from the danger of mass opposition, authoritarian regimes also face powerful urban elites with an interest in low food prices. Industrial interests can lower their costs by lobbying the government to lower food prices and cheapen labor and commodity inputs. Bureaucratic elites in developing autocracies often advocate policies which tax agriculture to fund industrialization projects.[26] As urbanization increases the number and concentration of these elites, their influence on policy under autocracy increases. Under democracy, on the other hand, elections increase the political salience of rural interests compared to urban interests because the widely dispersed agricultural population can influence politics via the ballot box with relative ease, while the power of city dwellers to affect policy through collective opposition is less salient than under autocracy. Urban interests are also often under-represented in electoral politics due to patterns of malapportionment which favor rural areas.[27] After controlling for the established development and comparative advantage patterns related to the size of the agricultural sector, therefore, the level of urbanization in a country does not have an effect on agricultural policy outcomes under democracy, while I predict that urbanization will be correlated with significantly lower levels of agricultural support under autocracy.

I hypothesize that income inequality under autocracy lowers the likelihood of collective action by food consumers, reduces their political power, and leads to higher food prices as compared to democracies. This is because, in contrast to inequality in landholdings, income inequality creates economic divisions among the population which decrease the likelihood of mass opposition to rural-biased policies. Following Engels' Law, the income elasticity of demand for food is less than one, so that richer households spend a smaller proportion of their income on food. Therefore, food price increases impose higher real costs on poorer households than richer households, and any potential benefits which could arise from collective action in opposition to higher food prices are lower for richer households than for poorer households, holding overall levels of income constant. Because this property of income inequality renders the benefits of political opposition to some consumers minimal, it decreases the likelihood of collective opposition against an authoritarian regime and its policies that increase food prices.[28] Where inequality is low, the population consists of a homogeneous group of poor citizens and has a higher likelihood of mobilization against food price policy than when inequality is high and the population contains a large group of richer citizens who have little concern for food prices. The effect of income inequality on policy under democracy is, on the other hand, to induce governments to implement lower food prices. Following the familiar Meltzer-Richard model, as the median voter's standard of living declines compared to that of the mean voter, pressures on democratic governments for redistribution to the pivotal voter increase.[29] Addressing the interests of poorer food consumers, democracies use agricultural policy to lower food prices and increase the real income of the median voter. Because income inequality decreases the power of food consumers to mobilize for lower food prices under autocracy and increases electoral pressures for lower food prices under democracy, it will be correlated with greater levels of agricultural support under autocracy compared to democracy, holding all else equal.

In stark contrast to the negative effects of income inequality on the ability of food consumers to organize in opposition to a regime, I argue that landholding inequality increases the political power of agricultural producers under autocracy and leads to greater levels of agricultural support, compared to democracies. At a fundamental level, this divergence in theoretical predictions is driven by the fact that income inequality creates divisions within a group of food consumers of a given size, while greater landholding inequality implies a smaller, richer, and more cohesive group of farmers. As landholding inequality increases, the number of agricultural producers declines, and they find it easier to solve the collective action problem and organize politically to pursue their interests.[30] Therefore, a higher level of landholding inequality should be associated with a greater political threat posed by landowning farmers. This dynamic holds more strongly under autocracies than under democracies, because landholding inequality implies redistributive pressures under democracy which mitigate its mobilization effects on farmers. Although democracies with smaller agricultural sectors support farmers more, on average, democracies have been found to tax agriculture at higher rates when landholding inequality is high, as they respond to the interests of the broader constituency of food consumers and redistribute resources to the median voter.[31] The distribution of land ownership, rather than the size of the agricultural sector, is also important under dictatorship because of the strong correlation between land inequality and authoritarianism found by empirical studies of the determinants of democratization.[32] Large landowners often take on an influential position within dictatorships, as hard-liners opposed to reform. In addition, the ownership of land, especially large farms, has long been associated with political influence under autocracy. Large landowners possess repressive resources, have traditionally held privileged positions in local political organizations and the church, and can use their position to control the political activities of their tenants and workers.[33] Because large landowners form a cohesive and powerful political elite under autocracy, while an unequal distribution of land stokes redistributive pressures under democracy, I argue that landholding inequality will be associated with greater levels of support for agriculture under autocracy compared to democracy, *ceteris paribus*.

The power of food consumers versus farmers under authoritarianism is determined by these structural factors, but operates through the mechanism of the threat of regime instability. It is the threat of unrest which renders highly urbanized consumers powerful and the threat of regime failure which renders large landowners powerful. Authoritarian leaders who implement agricultural policies in contradiction to the interests of powerful food consumers or producers should experience political instability, and they should respond to this instability by adjusting policy to favor powerful interests. These predictions are borne out by the experiences of authoritarian leaders in Egypt and Malaysia in the late 1960s and 1970s. In 1977, the Sadat dictatorship in Egypt followed advice from the International Monetary Fund to eliminate agricultural policies that had significantly reduced the price of some foods such as bread, flour, and cooking oil. Policies subsidizing basic consumption expenditure had been seen as a core element in the government's strategy of maintaining political stability, and increases in the prices of these staples resulted in food riots in major cities which posed a real threat to the regime. The government was forced to reverse its policy decision within days of its announcement and thereafter never questioned its commitment to a pro-consumer policy, guaranteeing a virtually limitless supply of low-cost bread to the urban population, even at the expense of decreasing producer prices.[34] Malaysia also experienced a period of intense political instability in the late 1960s, but one driven by rural interests rather than urban food consumers, and the Malaysian government responded to this instability by implementing an interventionist development policy which included significant price supports for farmers. After a contentious election in 1969, the ruling inter-ethnic coalition government was rocked by calls from powerful rural Malay politicians to move towards a single-party dictatorship under the United Malay National Organization (UMNO).[35] Rural Malays, especially the larger landowners who were powerful within UMNO, believed that the government had made too many policy concessions to non-Malay urban interests who exploited Malay rice farmers and profited from low prices for their produce.[36] In order to prevent a coup which could have driven the regime into single-party rule, Prime Minister Razak was forced to make concessions to these rural interests in the form of the New Economic Policy, a radical interventionist shift which significantly increased government support for the agricultural sector, and for rice farmers in particular.[37]

My account of agricultural policy formulation under authoritarian regimes generates three hypotheses on the relationship between structural measures of consumer and producer power and policy outcomes:

H1, Consumer Power: Higher levels of urbanization are correlated with lower levels of support to agriculture under authoritarian regimes versus democracies, *ceteris paribus*.

H1.1, Consumer Power: Higher levels of income inequality are correlated with higher levels of support to agriculture under authoritarian regimes versus democracies, *ceteris paribus*.

H2, Producer Power: Higher levels of landholding inequality are correlated with higher levels of support to agriculture under authoritarian regimes versus democracies, *ceteris paribus*.

Model and Data

In order to estimate the correlation between political power structures and agricultural policy, I specify fixed-effects panel regressions. Like all cross-national observational studies, this empirical analysis only approaches the identification of a causal relationship between my measures of consumer and producer power and policy outcomes. Levels of landholding inequality, income inequality, or urbanization are not randomly assigned to a given country in a given year as these socio-economic factors are co-determined by long-term development processes. Analysis of sub-national variation in agricultural policy could conceivably offer a better strategy for identification of the causes of policy changes. However, a study along these lines would be relatively uninformative and lack external validity because the policies which have the greatest effect on farmer and consumer welfare and which are comparable across countries—such as export restrictions and subsidies for inputs or consumption—are almost always implemented at the national level. For these reasons, my empirical strategy offers a good balance between methodological rigor and results which correspond to meaningful policy outcomes. I also go to some lengths to address problems of endogeneity in my models. To mitigate possible measurement error, I use the newest data available on global agricultural market distortions. I collected a new dataset on landholding inequality that maximizes the coverage and consistency of my data and use different model specifications to address concerns about its quality. To mitigate concerns about omitted variable bias, I follow previous research and include a full set of variables to control for relevant confounding factors in my models. Concerns about autoregression and autocorrelated errors have been mitigated by testing for autocorrelation using a Hausman test and specifying fixed-effects panel regressions with standard errors clustered by country. Problems of simultaneity, or a causal loop between policy and structural measures of consumer and producer power, are also mitigated by the fixed-effects regressions, which look only at changes in variables within countries across time, and by the fact that my measures of each group's power are unlikely to be affected by agricultural policy outcomes except in the very long run.

Dependent Variable

I use agricultural policy data collected in an international World Bank research project on agricultural market distortions between 1965 and 2010.[38] The coverage of these data is not evenly spread across space and time: developed democracies are conspicuously over-represented due to their importance to global agricultural trade. The former Soviet Union, socialist Eastern Europe, and the Middle East are notably under-represented due to their lack of comparable market institutions and significant food exports. However, the data are a great improvement over previous datasets, covering all regions, including countries which account for 92 percent of the global population and a wide variety of political systems.

The agricultural market distortions data estimate the direct and indirect effects of domestic government policy on the price incentives faced by farmers and food consumers. These policies include tariffs and trade measures, producer and consumer price distorting measures, exchange rate policy, distortions to intermediate input prices, and post-farmgate costs such as those imposed by state marketing monopolies.[39] The variables in the dataset measure policy by relating domestic prices to world market prices, summarizing aggregate national policy for all products for each country-year in measures which are production- and consumption-weighted to capture the total effects of agricultural policy within a given economy.

The key dependent variable, “Ag Support,” is a country's Nominal Rate of Assistance to total agriculture (NRA, nra_tott in the dataset). A country's total NRA is the production-weighted percentage by which domestic producer prices are above (or below if negative) the border price of like products. It should be noted that increases in agricultural support, as analyzed in my models, imply a corresponding increase in consumer prices. Governments almost invariably pass the benefits of support to farmers on to consumers in the form of consumer taxes, and the variables in my dataset measuring agricultural support and consumer taxes are correlated at 0.94. However, I also estimate my models taking both the Consumer Tax Equivalent (CTE, cte_covt in the dataset) and Relative Rate of Assistance (RRA, rra in the dataset) to agriculture as dependent variables. The results of these models are reported in Table 1 and are broadly in line with those which take the NRA as the dependent variable.

The NRA can logically vary between negative one (prices are reduced to zero) and any positive value (by which percentage prices are increased). For example, if a country's NRA value in a given year is 0.5, government policy increases producer prices by 50 percent compared to world market prices. If a country's NRA value in the same year is -0.5, it decreases producer prices by 50 percent. East Asian and European governments intervene very heavily in their agricultural sectors to increase prices for producers. Governments in developing regions such as South Asia and Africa tend to decrease farmers' produce prices.

Independent Variables

To measure the power of food consumers to threaten an authoritarian regime, I use the variable “Urban”: the proportion of a country's population living in cities, data which are available annually from the World Bank's World Development Indicators.[40] This variable ranged from 6 percent in Burkina Faso in 1968 to 91 percent in Australia in 2001. The mean level of urbanization in an authoritarian regime was 35 percent in my dataset, while in democracies the average was 66 percent.

To measure income inequality, I use the variable "Inequality," taken from the 2008 version of the Estimated Household Income Inequality dataset.[41] This dataset includes 3,419 Gini coefficients for 153 countries between 1963 and 2002, with a mean Gini of 0.39, a minimum of 0.20 and maximum of 0.64. Gini coefficients used in my analysis ranged from 0.26 in Sweden in 1980 to 0.58 in Mozambique in 1995. The mean level of inequality in an authoritarian regime was 0.44, while in democracies the average was 0.37.

To capture the power of agricultural producers to threaten authoritarian governments, I use a Gini coefficient "Land Gini," measuring the inequality of land ownership in a country. Data measuring land inequality are scarce; the Food and Agriculture Organization of the United Nations (FAO) has published national data since the 1950s, but coverage is sporadic in many developing countries. No single dataset has calculated all the landholding Gini coefficients available from the FAO data, therefore I make use of three datasets which utilize the same underlying data and maximize the number of observations I can include in my analysis.[42] I deal with the missing data problem in three ways. First, for countries with more than one observation between 1945 and 2000, I linearly interpolate them and perform an extrapolation them to fill out the missing country-years, censoring any resulting values which exceed the maximum found in the original data (0.98).[43] For countries with only one observation, I use that value for all country-years. I use the resulting data in a first set of analyses. Second, I take only data which are linearly interpolated between two values, excluding countries from my analysis which have only one land Gini datapoint. I use these data in a second analysis, reported in Table 3 in the Appendix.[44] Finally, I take the interpolated data and restrict them to the time period 1960–1990, when the coverage of the land inequality dataset is best.[45] I use the resulting data in a third analysis, also reported in Table 3.

Taking the first set of landholding Gini data, the mean landholding Gini coefficient was 0.60. The highest average levels of land inequality were to be found in Australia and New Zealand and South America where the average coefficients are 0.84 and 0.83, respectively. The lowest levels of land inequality in the dataset are in East Asia, with an average coefficient of 0.43. On average, the dictatorships included in subsequent analysis have slightly lower landholding Gini coefficients than democracies (0.57 compared to 0.62).

Regime Type Indicators and Control Variables

To test my hypotheses on the policy differences between democracies and authoritarian regimes, I interact my key explanatory variables with regime type indicators. First, I use a dummy variable created from the Polity dataset, which indicates whether a country's Polity score is below six, and thus not a democracy.[46] Second, I use a country's Polity 2 score as a continuous indicator, which ranges from -10 (fully institutionalized autocracy) to 10 (fully institutionalized democracy). In Table 4 in the Appendix, I list all the autocracies and democracies included in my analysis according to the binary Polity classification.

The models below also control for all major existing results in the literature on the political economy of agricultural policy. I include the natural log of per capita GDP ("Log GDP"), agriculture's share of GDP ("Ag in GDP"), arable land per capita, and agricultural land as a percentage of total land mass, with data from the World Bank World Development Indicators. I include a yearly international "Food Index" variable, which takes 2005 as a reference level (100) and is published by the World Bank in its regular commodity reports also known as “Pink Sheets.”[47] Summary statistics of all variables included in the models are shown in Table 2 in the Appendix.

Results

Results of four fixed-effects regressions, taking the annual NRA as dependent variable, are presented as Models 1–4 in Table 1. Two further fixed-effects models, which take consumer food taxes (CTE) and relative support to agriculture (RRA) as dependent variables, are presented as Models 5 and 6.[48] I direct the reader to Table 3, in the Appendix, for results of further models, which all broadly confirm the results depicted in Table 1.

Table 1: Fixed-Effects Models of Agricultural Policy HERE

Consumer Power

Turning first to the hypothesis H1—urbanization is an indicator of greater consumer power and will be correlated with lower levels of agricultural price supports under authoritarian regimes compared to democracy—Models 1, 2, 5, and 6 in Table 1 are supportive of this predicted relationship. Under democracies, urbanization is weakly positively correlated with agricultural supports, while under authoritarian governments this relationship is inverted and more urbanized polities are associated with significantly less support for farmers. This result holds in Model 1, which includes interactions between all threat variables and regime type, and in Model 2, which includes only an interaction between urbanization and regime type while including income and landholding inequality as controls. Models 5 and 6, which take consumer food taxes and relative levels of support to agriculture as independent variables, respectively, all show a robust negative correlation between urbanization and levels of support for farmers.

In Figure 1, I graph the substantive significance of the results of Model 1. The left-hand graph shows the predicted level of agricultural support (NRA) under both democracy and dictatorship across the range of the urbanization variable while holding all other variables at their means. Point predictions for each regime type at each level of urbanization are surrounded by their 95 percent confidence intervals. This graph clearly shows the divergent effects of urbanization on agricultural support under democracy, where support levels are increasing in urbanization, and dictatorship, where they are decreasing. The differences in support levels grow as urbanization increases, but due to high uncertainty around the point estimates at the ends of the urbanization distribution, they are only statistically significant at moderate levels of urbanization. The right-hand graph plots the marginal effect of dictatorship on agricultural support levels across the range of the urbanization variable. These results are more conclusive than the predictive margins shown in the left-hand graph. At low levels of urbanization, below 25 percent, the marginal effect of the binary indicator of dictatorship on agricultural support levels is positive, but not statistically significant. At levels of urbanization above 0.40, the marginal effect of regime type becomes significantly negative, while as urbanization levels reach the maximum levels observed in autocracies (83 percent), the average autocracy decreases agricultural produce prices by around 40 percentage points versus the average democracy, ceteris paribus.

Figure 1: Effects of Urbanization on Policy Outcomes by Regime Type

In Model 2, I include an interaction between urbanization and regime type while including landholding and income inequality as controls. These results only weakly confirm my hypothesis: higher urbanization is significantly correlated with lower support for farmers under dictatorship, but the size of the coefficient on the interaction terms is smaller and less statistically significant than in Model 1. The marginal effects of a change in regime type on the NRA are only around half the size of those in Model 1, for example. In Models 5 and 6, I test the effects of urbanization on consumer food taxes and support to agriculture relative to other sectors. I find similar results to Model 1; urbanization is only weakly correlated with policy outcomes under democracy, while under dictatorship countries shift to significantly lower levels of support for farmers as urbanization increases. In Table 3, in the Appendix, I show results of a model (3.1) which uses the Polity score as a continuous indicator of democratization, rather than a dummy variable for regime type. These results indicate that, although the coefficient on the interaction between Polity score and urbanization is only significant at the $p<0.08$ level, many democratic regimes do have greater levels of agricultural price support as urbanization increases. The random-effects Model 3.4, which includes fifteen regional dummy variables, also shows significantly lower levels of support for agriculture as urbanization increases under dictatorship.

These results are broad confirmation of the hypothesis that greater consumer power, as measured by urbanization, is negatively associated with agricultural producer supports under authoritarianism compared to democracy. They are also evidence that economic interest aggregation runs along strikingly different pathways under autocracy than democracy. Urbanization is not significantly correlated with agricultural price supports under democracy, even after controlling for development, political competition, and the size of the agricultural sector. Because democratic governments are responsive to electoral incentives, the concentration of the population in urban areas does not change their agricultural policy preferences.

Income inequality, on the other hand, is hypothesized in H1.1 to be an indicator of lower consumer power and to therefore be correlated with greater producer supports under autocracy versus democracy. Model results broadly support this hypothesis. Under democracy, I find evidence of a median-voter type effect of increasing inequality on agricultural support, as democratic governments respond to the interests of poor voters by lowering food prices. In all models reported in Table 1, under democracy, greater levels of income inequality are significantly correlated with lower agricultural producer supports, and thus also with lower food prices. Under autocracy, by contrast, greater levels of inequality are significantly correlated with increasing agricultural producer supports. The coefficient on the interaction term between the dictatorship dummy variables and the inequality variable is positive and statistically significant in Models 1, 3, 5, and 6. Table 3 in the Appendix presents the results of a further model (3.1), which uses the Polity score as a continuous indicator of democratization, rather than a dummy variable for dictatorship. The results of this model also indicate that more democratic regimes implement significantly lower levels of agricultural support as inequality increases, as the coefficient on the interaction between Polity score and inequality is negative and statistically significant. The random-effects Model 3.4 in Table 3 also indicates a negative and significant relationship between inequality and agricultural support under democracy and a positive and significant relationship under dictatorship.

In Figure 2, I illustrate the substantive importance of the relationship between income inequality and agricultural policy, graphing the results of Model 1. The left-hand graph shows the predicted level of agricultural support (NRA) under both democracy and dictatorship across the range of the inequality variable, while holding all other variables at their means. This clearly shows the opposing effects of income inequality on agricultural support levels across regime type. Under democracy, increases in inequality lead to large and significant decreases in agricultural support, and thus food taxes, from policies which increase produce prices by around 50 percent compared to world market prices to policies which have no significant effect on domestic producer incomes. Under dictatorship, increases in inequality are associated with relatively small but statistically significant increases in producer prices, of around 8 percentage points. The right-hand graph plots the marginal effect of dictatorship on agricultural support levels across the range of the inequality variable. These effects are large, negative, and statistically significant at low levels of income inequality but decreasing as inequality increases and positive at very high levels of inequality. Where the income Gini coefficient is only 0.30, authoritarian governments provide policies which are 39 percent points lower vis-à-vis world market prices than democracies, on average. Above the average levels of inequality observed in dictatorships (0.44), however, authoritarian regimes are no longer predicted to follow significantly more consumer-friendly policies than democracies.

Figure 2: Effects of Inequality on Policy Outcomes by Regime Type HERE

The results of these models provide support for the hypothesis laid out in H1.1 and still more evidence that agricultural policy-making runs along dramatically different lines under autocracy than democracy. Democratic governments follow policies that work against income inequality, lowering food prices as a form of redistribution to poorer voters. Autocratic governments, on the other hand, follow the opposite policy and respond to consumer interests only when they are backed by an acute threat to regime stability. Because consumers divided by inequality are less likely to pose a collective threat to a regime, they have less political power to oppose agricultural policies which run counter to their interests, and authoritarian governments significantly increase support to agriculture as inequality increases.

Producer Power

I turn now to the relationship between producer power, as measured by landholding inequality, and agricultural policy as hypothesized in H2. Because landholding inequality implies a smaller and more powerful group of agricultural producers, which can more effectively threaten the stability of an authoritarian regime, I predict that this variable will be associated with higher levels of agricultural producer support under autocracy compared to democracy, on average.

The results of four models presented in Table 1, which use my full landholding inequality dataset, all provide support for this hypothesis. Under democracy, landholding inequality is only weakly correlated with agricultural policy outcomes, and the sign on the uninteracted coefficient on “Land Gini” is not consistent across models. Models reported in Table 3 in the Appendix also find no robust correlation between landholding inequality and agricultural policy under democracy. Interestingly, this result is not consistent with that of Olper, who finds that democracies implement lower levels of support to agriculture as land inequality increases. My results suggest that the ability of farmers to lobby for advantageous regulation under democracy depends more on the size of the sector and its comparative advantage than on average landholding size.[49]

In my models, the landholding inequality variable is only robustly correlated with policy outcomes when interacted with indicators of political regime type. The coefficient on the interaction between “Land Gini” and the dummy variable for dictatorship is positive and statistically significant in Models 1, 4, 5, and 6, although the level of statistical significance in Model 4, which includes income inequality and urbanization as controls, is lower at $p<0.052$. As levels of landholding inequality increase, authoritarian regimes support agriculture at significantly higher levels compared to democracies. Because the landholding inequality dataset is subject to substantial missingness, I estimate two more models using only a subset of the data. These models are reported as Models 3.2 and 3.3 in Table 3 in the Appendix and include uninteracted urbanization and inequality variables as controls.[50] Their results are in line with the findings based on the full dataset presented in Table 1. Model 3.2 uses only data interpolated between two observations and excludes all extrapolated data and countries with only one observation. This model indicates a positive correlation between landholding inequality and agricultural support under authoritarianism, though the significance of the coefficient is lower, at $p<0.056$. Model 3.3, like Model 3.2, only uses interpolated data but further restricts the sample to the years 1960–1990 when the number of land inequality observations is densest. The coefficient on the interaction term between landholding inequality and the dictatorship dummy is also positive and statistically significant in this model. Model 3.4, a random-effects model including fifteen regional dummies, generates results very similar to those in Table 1, with landholding inequality significantly positively correlated with support for agriculture under dictatorship, and negatively but insignificantly correlated with support under democracy.

Figure 3: Effects of Landholding Inequality on Policy Outcomes by Regime Type HERE

Figure 3 illustrates the substantive importance of the relationship between regime type, landholding inequality, and agricultural policy outcomes, graphing results from Model 1. The left-hand graph shows the predicted level of agricultural support (NRA) under both democracy and dictatorship across the range of the landholding inequality variable, while holding all other variables at their means. Predicted levels of agricultural support are flat under democracy, but increasing under dictatorship. Differences in predicted policy outcomes are large, with democracies supporting agriculture around 40 percentage points more at very low levels of landholding inequality, but decreasing as landholding inequality increases. The model predicts statistically significant differences in policy outcomes across a relatively large range of land Ginis, from 0.45 to 0.60, above which autocracies are no longer predicted to implement significantly lower levels of producer support than democracies. The right-hand graph plots the marginal effect of dictatorship on agricultural support levels across the range of the “Land Gini” variable. This effect is decreasing in landholding inequality. At a landholding inequality Gini of 0.4, around one standard deviation below the mean level for autocracies in my dataset, autocracies are predicted to support agriculture 37 percentage points less than democracies vis-à-vis world market prices, on average. At autocracies' average level of landholding inequality (0.57), democracies still provide a level of support that is significantly higher than autocracies, though the difference is only around 20 percentage points. However, as landholding inequality increases to high levels (0.725, the maximum in the dataset is 0.98), there is no statistically significant policy difference between regime types, ceteris paribus.

These results provide robust support for hypothesis H2: greater levels of landholding inequality are correlated with higher levels of support to agriculture under authoritarian regimes versus democracies, ceteris paribus. Landholding inequality is an indicator of the latent power of agricultural producers to influence economic policy-making under authoritarianism, and as landholding inequality increases, the tendency for authoritarian regimes to support agriculture less than democracies is significantly mitigated and even eliminated at very high levels of landholding inequality.

Conclusion

In this article, I revise one-dimensional conceptions of agricultural policy under authoritarianism, which argue that all autocracies subsidize farmers significantly less than democracies. Authoritarian leaders make agricultural policy not by weighing up campaign contributions and votes as in democracies, but by making a grim calculation of the relative threats to their regime posed by producers and consumers. Agricultural policy-making under authoritarianism can be explained as a response to these political threats. In some cases, for example when land inequality is very high, urbanization very low, or inequality is high, farm policy is no more consumer-biased under dictatorship than under democracy. On the other hand, when land or income inequality is very low or urbanization is high, dictatorships are more likely to tax agriculture, as established by previous studies.

Given the considerable impact of agricultural market distortions on global trade and development, enhancing our understanding of the political determinants of distortions to global agricultural markets is an important step towards eliminating welfare-reducing policies. Revealing the full spectrum of authoritarian agricultural policies also provides a deeper understanding of the determinants of rural poverty, rural-urban migration and the associated economic inequality, and political instability. Finally, accounting for policies like those used to support food producers or consumers is an important amendment to economic theories of democratization, which draw one-way lines of causality from socio-economic structures to political outcomes. By exploring the extent to which development and structural change are themselves shaped by authoritarian regimes, democratization scholars can open up an important avenue of research, which will deepen our understanding of the determinants of authoritarian durability and democratization.

Table 1 Fixed-Effects Models of Agricultural Policy

	(1) NRA	(2) NRA	(3) NRA	(4) NRA	(5) CTE	(6) RRA
Dict	-1.26***	0.13	-1.06***	-0.45***	-1.46***	-1.17***
	(0.22)	(0.08)	(0.28)	(0.20)	(0.34)	(0.26)
Urban	0.46	0.63	0.32	0.20	0.39	0.84
	(0.45)	(0.66)	(0.57)	(0.54)	(0.54)	(0.55)
Dict*Urban	-0.64***	-0.35**			-0.65**	-0.82***
	(0.18)	(0.17)			(0.25)	(0.21)
Land Gini	-0.19	0.64	0.56	0.29	-0.29	0.05
	(0.65)	(0.65)	(0.60)	(0.61)	(0.85)	(0.74)
Dict*Land G	0.96***			0.64*	1.22***	1.17***
	(0.18)			(0.32)	(0.23)	(0.20)
Inequality	-1.89***	-0.76*	-2.03**	-0.84**	-1.78**	-1.65**
	(0.45)	(0.38)	(0.65)	(0.37)	(0.70)	(0.53)
Dict*Ineq	2.12***		2.37***		2.25**	1.90***
	(0.40)		(0.62)		(0.65)	(0.51)
Log GDP/Cap	-0.02	-0.01	0.02	0.01	0.08	0.04
	(0.07)	(0.10)	(0.08)	(0.08)	(0.10)	(0.08)
Ag in GDP	-0.56***	-0.59*	-0.58*	-0.58**	-0.02	-1.03**
	(0.28)	(0.32)	(0.29)	(0.28)	(0.32)	(0.32)
Food Index	-0.00***	-0.00***	-0.00***	-0.00***	-0.00***	-0.00***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Observations	1326	1326	1326	1326	1327	1291
AIC	-695.24	-608.75	-641.78	-624.71	-85.24	-466.05
Countries	56	56	56	56	56	53

Standard errors in parentheses. All models with fixed effects, clustered standard errors.

* p<0.10, ** p<0.05, *** p<0.001.

Figure 1 Effects of Urbanization on Policy Outcomes by Regime Type

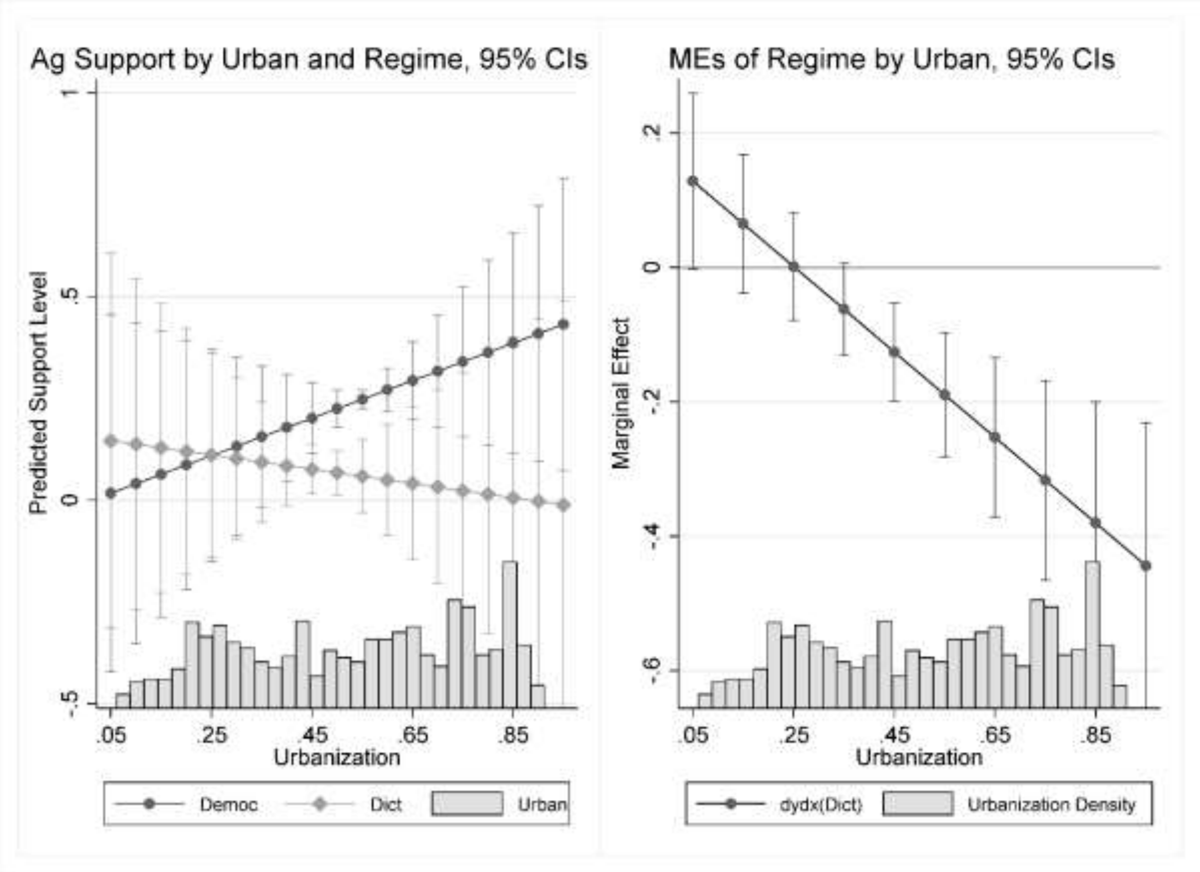


Figure 2 Effects of Inequality on Policy Outcomes by Regime Type

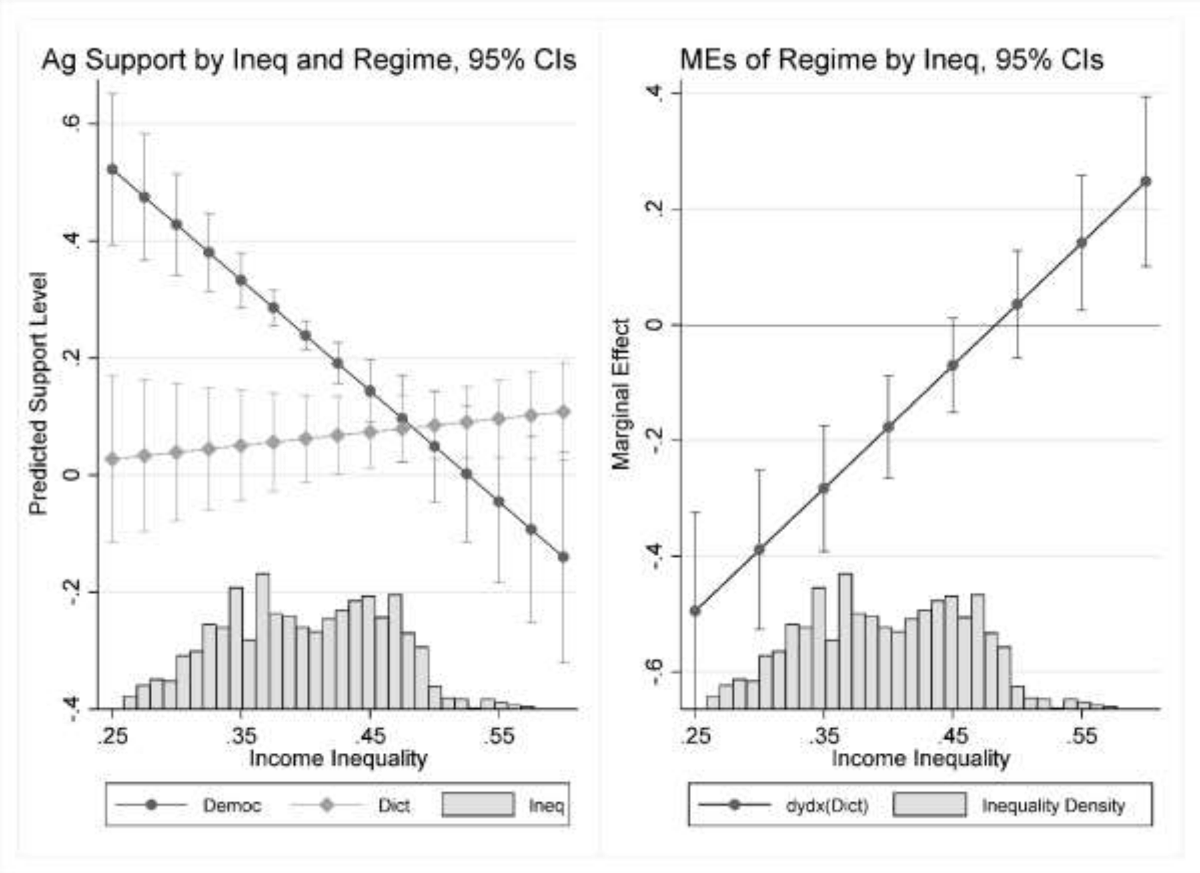
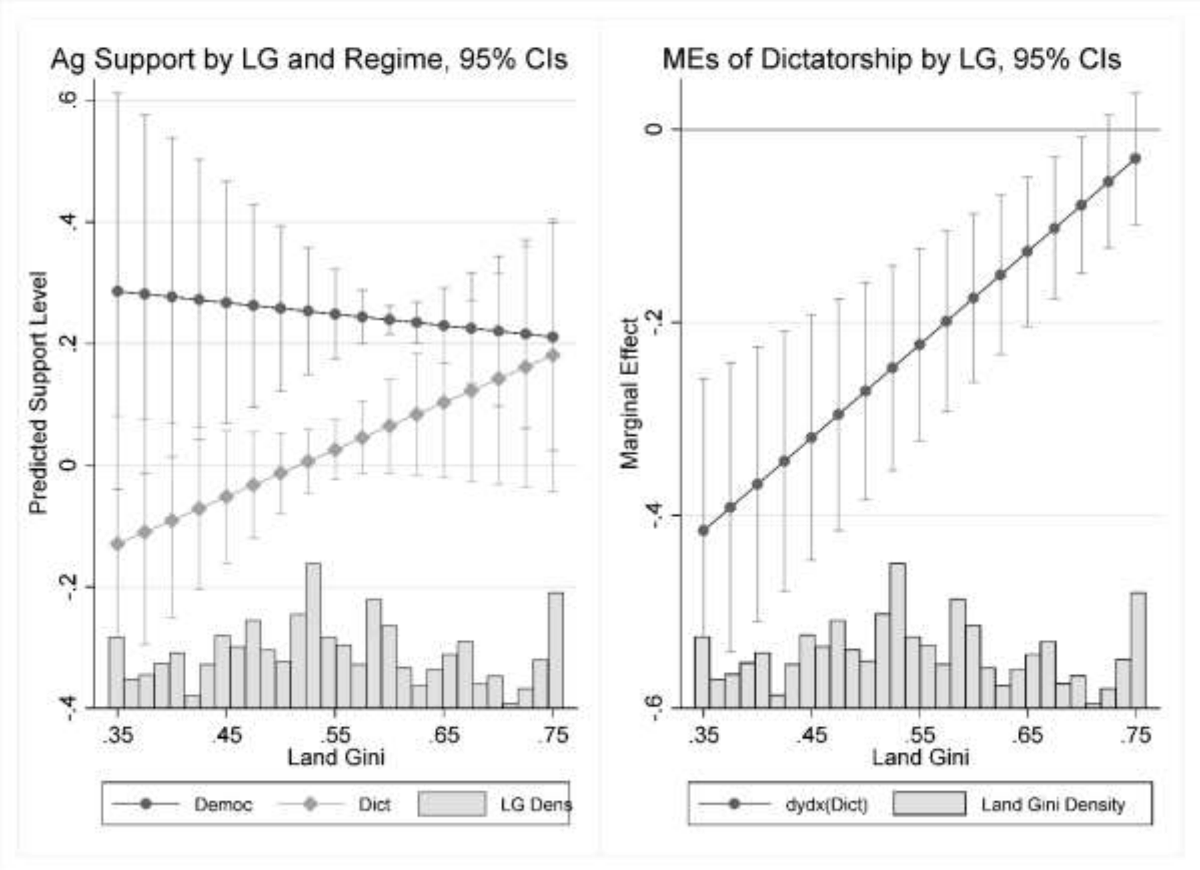


Figure 3 Effects of Landholding Inequality on Policy Outcomes by Regime Type



NOTES

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[44] Due to space constraints, the Appendix is not in the print version of this article. It can be viewed in the online version, at www.ingentaconnect.com/cuny/cp.

[45] 169 of a total 230 land Gini observations are from this period.

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[49] Olper, 2007.

[50] Including the interaction terms strengthens my results, but I omit them to allay concerns that the correlation between landholding inequality and policy is driven by the inclusion of higher-order interaction terms.